

6-2 Multiplying and Dividing Radicals

Warm Up

1) $\sqrt{75}$

$$\sqrt{25 \cdot 3}$$

$$5\sqrt{3}$$

2) $\sqrt[2]{12x^3}$

$$\sqrt{4 \cdot 3 \cdot x^2 \cdot x^1}$$

$$2x\sqrt{3x}$$

3) $\sqrt[3]{27x^{18}}$

$$3x^6$$

4) $2\sqrt[3]{16x^5y^8}$

$$4x^1y^2\sqrt[3]{2x^2y^2}$$

Review of some Exponent Rules

Division		
• If bases the same subtract the exponents	$\frac{x^a}{x^b} = x^{a-b}$ or x^{b-a}	

I. Multiplying Radicals

The roots (indexes) must be the same to multiply radicals!
If not same, you can't do it.

same
 $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$

$$\sqrt[3]{5} \cdot \sqrt[3]{6} = \sqrt[3]{30}$$

Multiply number to number, radical to radical.

After multiplying, always check to see if you can simplify.

$$\sqrt{5} \cdot \sqrt[3]{6} = \sqrt[1]{8} \sqrt[27]{1}$$

Ex1) $2\sqrt[3]{4x} \cdot 5\sqrt[3]{6x^2}$

Ex2) $6\sqrt{2x^4} \cdot 5\sqrt[3]{2x}$

$$\begin{aligned} & 10\sqrt[3]{24x^3} \\ & 10\sqrt[3]{8 \cdot 3 \cdot x^3} \\ & 20x\sqrt{3} \end{aligned}$$

multiply #s and simplify. Can't mult. the radicals

$$30x^2\sqrt{2} \cdot \sqrt[3]{2x}$$

$$\frac{1^3}{2^3} = \frac{1}{8}$$

YOU!

1. $(2\sqrt{10})(3\sqrt{5})$

$$\begin{aligned} & 6\sqrt{50} \\ & 6\sqrt{25 \cdot 2} \\ & 30\sqrt{2} \end{aligned}$$

2. $5\sqrt{4x^5y^3} \cdot 4\sqrt{2xy^4}$

$$\begin{aligned} & 20\sqrt[5]{8x^4y^7} \\ & 40x^3y^3\sqrt{2y} \end{aligned}$$

3. $\sqrt[3]{72x^3y^2} \cdot \sqrt[3]{xy^3}$

$$\begin{aligned} & \sqrt[3]{8 \cdot 9x^4y^5} \\ & 2xy\sqrt[3]{9xy^2} \end{aligned}$$

II Dividing

The roots (indexes) must be the same to divide radicals!
If not same, you can't do it.

- With dividing, sometimes it's useful to see the whole fraction under one radical, and sometimes it's easier to have them separated.

Rules: $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$ or $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$ (as long as $b \neq 0$)

Ex3. $\frac{\sqrt[3]{15y^5}}{\sqrt[3]{3y^2}}$ think $\sqrt[3]{\frac{15y^5}{3y^2}} = y\sqrt[3]{5}$

Ex 4. $\sqrt{\frac{25xy^2}{16}}$ think $= \frac{\sqrt{25xy^2}}{\sqrt{16}} = \frac{5y\sqrt{x}}{4}$

4. $\frac{\sqrt{18x^5}}{\sqrt{2x^3}}$ $\times$ $5-3$
 $= \sqrt{9x^2} = 3x$

5. $\frac{\sqrt{56x^5y^5}}{\sqrt{7xy}}$
 $= \sqrt{8x^4y^4} = 2x^2y^2\sqrt{2}$

III. "Rationalize the denominator" is the process of writing an equivalent expression with no radicals in the denominator. Final answers should have NO RADICALS IN THE DENOMINATOR

- Multiply by whatever it takes to undo the radical, remember:

$\sqrt[n]{a^n} = a$

(You want to make the power on each factor in the radicand to equal the root)

Ex5. $\frac{7x}{\sqrt{3}}$ $\times \frac{\sqrt{3}}{\sqrt{3}} = \frac{7x\sqrt{3}}{3}$

7. $\frac{1}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5}$

8. $\frac{3}{2\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{3\sqrt{5}}{10}$

$\sqrt{3^2} = 3$

$\frac{1}{\sqrt{25}} = \frac{1}{5}$